

Chapter 46 — Neuromuscular Dysfunction of the Lower Urinary Tract in Children

- **Definition:** neuromuscular (neurogenic, neuropathic) lower urinary tract (LUT) dysfunction is **bladder and/or sphincter dysfunction from aberrant innervation**. It may be congenital or acquired, central or peripheral, transient or permanent.
- **Commonest congenital cause: neural tube defect (NTD)**, used almost synonymously with spina bifida (SB).
- **Survival has transformed:**
 - infant survival rose from **under 10% before 1960 to over 85% by the mid-1990s**;
 - **first-year mortality is now about 5%**;
 - **about 90% survive to 10 years**.
- **Pivotal urological advance: clean intermittent catheterisation (CIC), Lapides (1971)**. It prevents upper-tract deterioration.
- **Predictors of mortality:**
 - long-term mortality **rises with the level of neurological deficit**;
 - the **strongest infant risk factors are preterm birth (relative risk [RR] about 4.5) and low birth weight (RR about 4.8)**.
- **As survival lengthens, renal failure becomes a leading cause of morbidity and death**. This is why the urinary tract needs lifelong surveillance from infancy.
- **About 10% of deaths in SB are urological**.
- **The urologist protects the urinary tract and enables sexual function**.

TABLE 46.1 CAUSES OF NEUROMUSCULAR LOWER URINARY TRACT DYSFUNCTION

Congenital	Acquired
Neural tube defect	Extensive pelvic surgery
Occult NTD: lipomeningocele, other spinal dysraphisms	Central nervous system (CNS): cerebral palsy; brain tumours, infarcts, encephalopathies
Sacral agenesis	Spinal cord: trauma; transverse myelitis
Anorectal malformations	

- **The table is the chapter's skeleton:** NTD first, then the other causes grouped by brain, spinal cord and pelvis.
- **Trap:** in anorectal malformation, the LUT dysfunction usually comes from an associated cord lesion, not from the operation.

NEURAL TUBE DEFECTS

Background and Classification

- **NTDs are abnormal development of the spinal canal and cord**. They are classified by **location, severity (open or closed) and the structures involved**.

NTD CLASSIFICATION — LOCATION AND SEVERITY

Type	Site	Open / closed	Defining feature
Myelomeningocele	Spinal	Open	Cord, roots and meninges protrude; the most severe and the most common open NTD; usually lumbosacral
Meningocele	Spinal	Usually skin-covered	Meninges only; the cord stays in the canal
SB occulta	Spinal	Closed	Mildest; up to 15% of the population; skin-covered; usually a non-pathological variant
Sacral agenesis (caudal regression)	Spinal	Closed	Absent sacral vertebrae; may coexist with tethering
Anencephaly	Cranial	Open	Most of the brain and skull absent; very limited survival
Encephalocele	Cranial	Open	Brain and meninges herniate through a skull defect

- **Open versus closed:**
 - an **open NTD** exposes neural tissue;
 - a **closed NTD** is skin-covered.
- **Prenatal detection: maternal α -fetoprotein and ultrasound.** This allows counselling and planning of delivery.

Epidemiology

- **Spinal closure:**
 - begins **around day 18**;
 - proceeds **caudally from the cephalad end**;
 - is **complete by day 35**.
 - The mechanism of failure is **unknown and multifactorial**.
- **Worldwide prevalence is about 2 per 1000 births**, concentrated in **developing countries**.
- **In the USA it has plateaued at about 30 per 100,000**, lowest in non-Hispanic Black children. It varies by place and over time.

Risk Factors for the Development of Neural Tube Defects

- **Folate is central.** Risk rises with **low folate, impaired folate pathways or anti-folate antibodies**.
- **Evidence for folic acid:**
 - **supplementation cuts NTDs by about 50–70%** (randomised trials);
 - **flour fortification (late 1990s) cuts them by about 20–50%**. Fortification was introduced because only one-third of women supplement.
- **US Public Health Service (1992): 400 μ g daily for all women of child-bearing age.**
 - **Why all women:** the tube closes before most women know they are pregnant.
 - **Optimal timing:** at least 4 weeks before conception and through the first month.
- **Genetics:**
 - **monozygotic concordance is higher than dizygotic**;
 - **recurrence is 5% after one affected pregnancy and no more than 10% after two, so not monogenic**;
 - **no clinically actionable human gene has been found**.

- **Other maternal risk factors:**

- age at either extreme;
- obesity, diabetes, fever or influenza, caffeine;
- occupational, hydrocarbon and pesticide exposure;
- low socioeconomic status, passive smoking, periconceptional stress, low weight gain;
- folate antagonists (valproate, carbamazepine);
- prior miscarriage or birth defect; higher birth order.

Pathogenesis

- **Almost every open NTD has an Arnold–Chiari II malformation.**
 - **Components:** hindbrain herniation, brainstem anomalies, low venous sinuses, a small posterior fossa.
 - **Associated with:** hydrocephalus, and motor, cranial-nerve and cognitive deficits.
 - It carries mortality even after surgery.
- **Hydrocephalus treatment:**
 - traditional: **ventriculoperitoneal (VP) shunt**;
 - alternative: **endoscopic third ventriculostomy with choroid plexus cauterisation**, which controls it in **over 70% without a shunt**, with **equivalent neurocognitive outcome**.
- **The myelomeningocele lesion varies** with which neural elements evert into the sac.
- **The bony level differs from the neurological level by 1–3 vertebrae either way. LUT function cannot be predicted from the bony defect or from leg function.**

Perinatal Concerns.

- **The injury progresses in utero:** leg movement is lost and hindbrain herniation and hydrocephalus worsen.
- **Prenatal coverage preserves function** in animal models.
- **"Two-hit" theory:** failure of tube formation, then **ongoing mechanical and chemical injury to exposed tissue**. It follows that **in-utero repair should help**.

Management of Myelomeningocele Study (MOMS) — landmark randomised trial

- **Design: prenatal repair before 26 weeks** versus postnatal repair.
- **Benefits of prenatal repair:**
 - fetal or neonatal death or shunt need: RR 0.70, yet **40% still needed a shunt**;
 - **better cognitive and motor composite at 30 months**.
- **Neutral results: no maternal deaths; no difference in fetal or neonatal deaths.**
- **Costs of prenatal repair:**
 - **more obstetric complications:** oligohydramnios, chorioamniotic separation, abruption, transfusion, membrane rupture;
 - **one-third had uterine-scar dehiscence or thinning**;
 - **delivery at 34.1 versus 37.3 weeks; 13% before 30 weeks**;
 - more re-tethering procedures, more respiratory distress, lower birth weight.
- **Since MOMS, open and fetoscopic refinements have reduced maternal and fetal risk.**

TABLE 46.2 SPINAL LEVEL OF MYELOMENINGOCELE

Cervical-high thoracic	Low thoracic	Lumbar	Lumbosacral	Sacral
2%	5%	26%	47%	20%

- The pattern: two-thirds are lumbosacral or sacral, and over 90% including lumbar. High lesions are rare.
- Trap: the level still does not predict bladder function.

KEY POINTS: NEURAL TUBE DEFECTS

- NTD is the commonest cause of neuropathic LUT dysfunction in children; **myelomeningocele is the most severe and the most common type.**
- Folic acid is the key prevention, from ≥ 4 weeks before conception through the first month.
- LUT effects cannot be predicted from the bony defect, the neurological level or leg function.
- About 10% of SB deaths are urological.

Concept — The spinal lesion in SB is **both unpredictable and progressive**: the bony level misstates the neurological level, and the exposed cord keeps being injured in utero. Every later principle follows: **urodynamic definition rather than anatomical inference, early and repeated study, and lifelong surveillance.**

NEUROPATHIC LOWER URINARY TRACT DYSFUNCTION

Classification

- Normal function:
 - storage at low pressure;
 - emptying by a detrusor contraction coordinated with a fall in outlet resistance.
- Denervation disrupts this, producing storage failure, emptying failure, or both.
- Terminology: "overactive" replaces hyperreflexic; "underactive" replaces hypo- or areflexic.

NEUROGENIC DETRUSOR OVERACTIVITY VERSUS NEUROGENIC UNDERACTIVE BLADDER

Neurogenic detrusor overactivity (NDO)	Neurogenic underactive bladder
Involuntary, frequent contractions; if sensate → urgency and frequency	Impaired detrusor stimulation → ineffective contraction
Storage lost through small capacity and high pressure, from pathological detrusor remodelling (multifactorial)	Incomplete emptying, retention, overflow incontinence, urinary tract infection (UTI)

- Incontinence occurs with or without sensation.
- Low outlet resistance also impairs storage. It results from an untuned bladder neck and/or a denervated external sphincter.

Concept — Every neuropathic pattern reduces to **two components: the detrusor (overactive or underactive) and the outlet (competent, incompetent or dyssynergic)**. Their combination sets both the continence problem and the renal risk.

Assessment

- Goals of assessment: preserve renal function, achieve continence, prevent UTI, enable sexual function. Each depends on reliable and timely testing.

- **Four bodies publish guidance, broadly in agreement:**
 - **International Children's Continence Society (ICCS, 2012);**
 - **British Association of Paediatric Urologists (BAPU, 2016);**
 - **Spina Bifida Association (SBA, 2018);**
 - **European Association of Urology (EAU, 2022).**

TABLE 46.3 ASSESSMENT OF NEUROGENIC LUT DYSFUNCTION

Test	ICCS (2012)	BAPU (2016)	SBA (2018)	EAU (2022)
Urodynamics (UDS)	After recovery from SB surgery; before untethering; ≥6 weeks after spinal cord injury (SCI); yearly in toddler years; 2–3 months after starting or changing an antimuscarinic	"Select few" on clinical or radiological grounds; if stable on CIC, only if indicated	Within 3 months; annually to 36 months; repeat for UTI, upper-tract or continence change	Within 3 months; then annually as indicated
Ultrasound	Initial; 6, 12, 18, 24 months; yearly or twice-yearly after the toddler years; yearly in adolescence; every 2 years after the growth spurt; every 3 years in adulthood	Yearly	After birth; 6, 12, 18, 24 months; annually after 24 months; repeat for UTI or worse UDS	After birth; annually
Dimercaptosuccinic acid (DMSA)	—	—	—	Baseline
Cystography (voiding cystourethrogram [VCUG] or video UDS)	If hydronephrosis, ureteric dilatation, renal size discrepancy, wall thickening or hostile dynamics; repeat for vesicoureteric reflux (VUR), new dilatation, parenchymal loss or febrile UTI	At initial evaluation	—	"May be helpful"
Laboratory	Creatinine after the first week, if indicated	—	Creatinine within 3 months; chemistries at 5 years, then yearly creatinine; yearly chemistries and B12 from 2 years after reconstruction	Creatinine in the first week; cystatin C more accurate in older children; yearly B12 from 2 years after reconstruction

- **The pattern:**
 - **ICCS, SBA and EAU agree on an early study within 3 months and serial ultrasound through the first 2 years;**
 - **BAPU is the most selective.**
- **The traps:**
 - **only ICCS and BAPU positively recommend cystography;**
 - **baseline creatinine is recommended by three bodies, but not BAPU;**

- **B12 surveillance starts 2 years after bowel reconstruction** (time for terminal-ileal loss to deplete stores).

History and Physical Examination

- **Without NTD stigmata, symptoms overlap with non-neurogenic dysfunction** (incontinence, LUT symptoms [LUTS], UTI).
- **Red flags: worsening incontinence or LUTS, recurrent febrile UTI, upper-tract change.**
- **Examine the spine and legs in every child.**
- **Cutaneous markers are present in about 90% of occult dysraphism:**
 - hair tuft;
 - dermal vascular malformation;
 - sacral dimple;
 - abnormal gluteal cleft.
- **Mild sacral agenesis: the child may walk; look for a short gluteal cleft and flat buttocks.**
- **Assess sexual function in adolescents.**

Urodynamics

- **Urodynamics are the primary test driving management.**
- **Landmark — McGuire (1981): detrusor leak point pressure (DLPP) >40 cm H₂O predicts VUR and upper-tract change.**
- **Poor compliance (volume ÷ end-filling pressure) also predicts renal injury.**
- **No exact injurious threshold is known, but high storage pressure harms the kidney.**

TIMING OF URODYNAMICS BY CONDITION

Setting	When
Open NTD	Once transportable and able to lie supine; usually within the first few months; safe from 3 months
Closed NTD or cord tumour	Before neurosurgery and 3–4 months after
SCI	After spinal shock resolves (about 3–4 months)

- **Newborn myelomeningocele:**
 - **most have detrusor contractions**, even with high lesions that spare the sacral cord;
 - **over a third are acontractile**, with good or poor compliance.
- **Interrater reliability is poor.**
- **Detrusor–sphincter dyssynergia (DSD):**
 - **definition: failure of the sphincter to relax during a voiding or overactive contraction;**
 - **significance: a presumed driver of pathological remodelling**, so accurate diagnosis matters;
 - **diagnosis: electromyography (EMG); bladder-neck fluoroscopy is unreliable.**

SURFACE PATCH EMG VERSUS DIRECT CONCENTRIC NEEDLE EMG

Surface patch EMG	Direct concentric needle EMG
Commonest; perineal electrodes	Reliable and objective; placed with the UDS catheters (girls: 3 or 9 o'clock beside the meatus; boys: transperineal)
	Needs specialist training or a neurologist

Surface patch EMG**Direct concentric needle EMG**

Unreliable in UMPIRE: DSD not supported by other UDS elements → **removed from the "high-risk" category**

Coordination only

Also shows **acute versus chronic denervation**; clarifies suspected re-tethering when postoperative scarring makes MRI equivocal

- **UMPIRE** is the Centers for Disease Control and Prevention (CDC) Urologic and Renal Protocol for the Newborn and Young Child with SB, a landmark prospective protocol.
- **Needle EMG in newborn myelomeningocele** gives a baseline for comparison:
 - intact sacral arc about 40%;
 - partial denervation about 25%;
 - complete denervation about 35%.

URODYNAMIC RISK CATEGORIES FOR RENAL DETERIORATION

Category	Definition	UMPIRE revision
Hostile	End-filling pressure or DLPP ≥ 40 cm H ₂ O, or NDO with DSD	DSD removed
Intermediate	NDO, reduced compliance, pressure 25–39 cm H ₂ O	—
Abnormal but safe	Pressure < 25 cm H ₂ O	—
Normal	Normal capacity; compliance > 15 mL/cm H ₂ O; no NDO or DSD; minimal residual	—

- **Needed:** non-invasive or ambulatory urodynamics, and machine-learning interpretation. Both depend on **standardised data that do not yet exist**.

Imaging*Ultrasound.*

- **Ultrasound is non-invasive and tracks renal size, growth and dilatation, which predicts renal health.**
- **It is unreliable for scarring, VUR, and bladder wall thickness.**
- **Baseline scan:** before newborn discharge; soon after injury in acquired causes.
- **UMPIRE baseline:**
 - 56% normal;
 - 40% mild hydronephrosis;
 - 4% moderate to severe.
- **It also measures residual urine, which guides the need for CIC.**
- **Interval:** yearly, or 6-monthly for 2 years then yearly.
- **Spinal ultrasound works before the posterior arches ossify (first few months)** to detect occult dysraphism.

Dimercaptosuccinic Acid Renal Scintigraphy.

- **Technetium-99m DMSA is the most sensitive test for cortical defects.**
- **It is used inconsistently** because of isotope availability and expertise.
 - **UMPIRE:** only one-third scanned; about 8% had defects.

- **EAU recommends a baseline DMSA** for longitudinal assessment.

Cystography.

- **Cystography is fundamental; ideally video UDS**, which links anatomy to pressure. **VCUG otherwise.**
- **Main purpose: VUR**, especially when there is:
 - **on ultrasound:** dilatation, renal discrepancy or wall thickening;
 - **on UDS:** overactivity, poor compliance, high leak point pressure or DSD.
- **UMPIRE: grade III–V VUR in only 7% of newborns.**
- **High-grade VUR acts as a pressure pop-off**, so urodynamic pressures become unreliable.
- **The baseline battery:**
 - gives a **reference** for later comparison;
 - **detects deterioration or progressive denervation early;**
 - **identifies at-risk infants before upper-tract change**, so prophylaxis can start;
 - **informs counselling.**

Laboratory Studies

SERUM CREATININE VERSUS SERUM CYSTATIN C IN SB

Serum creatinine	Serum cystatin C
Made from muscle creatine; filtered and secreted	13-kDa cysteine proteinase inhibitor made at a constant rate by all nucleated cells; filtered, not secreted, catabolised in the proximal tubule
Depends on age, sex, height and above all muscle mass	Independent of muscle mass
Low muscle mass → normal creatinine despite a falling glomerular filtration rate (GFR)	Tracks GFR better than the Schwartz formula; $\log(\text{GFR}) = 1.962 + [1.123 \times \log(1/\text{cystatin C})]$
Cheap and consistent → baseline recommended	Less reliable in early impairment; limited availability; affected by inflammation and protein catabolism

- **Gold standard for GFR: a nuclear-medicine clearance study.** It is invasive and involves radiation.
- In adults, cystatin C predicts end-stage renal disease (ESRD) and death better than creatinine.
- If cystatin C is unavailable, or mild impairment is suspected → clearance study.

KEY POINTS: NEUROPATHIC LUT DYSFUNCTION: ASSESSMENT

- **Suspect neuropathy with worsening incontinence or LUTS, recurrent febrile UTI or upper-tract change.**
- **Urodynamics drive management despite poor interrater reliability.**
- **DSD is a presumed primary driver of remodelling.**
- **SB ultrasound: 56% normal, 40% mild, 4% moderate to severe dilatation.**
- **Cystography is fundamental, ideally video UDS; about 7% have dilating VUR.**
- **Baseline creatinine is recommended.**

Concept — Assessment rests on one causal chain, **high storage pressure → upper-tract injury**. Each test measures a different link: urodynamics the pressure, EMG the outlet that generates it, cystography the

reflux that transmits it, ultrasound and DMSA the damage, and cystatin C what low muscle mass hides from creatinine.

Management

- **Priorities, in order: kidneys first**, then continence, UTI avoidance, and sexual function and fertility.

TABLE 46.4 MANAGEMENT OF NEUROGENIC LUT DYSFUNCTION

Domain	ICCS (2012)	BAPU (2016)	SBA (2018)	EAU (2022)
CIC	After SB surgery if not emptying ; for high-grade VUR or NDO ; as soon as feasible after SCI	No consensus	First year for mixed incontinence if ultrasound or UDS indicates; later for upper-tract change, recurrent UTI or hostile UDS; child involved from 3–5 years	Suggest early ; may delay if the sphincter is underactive or there is no DSD
Drugs	Antimuscarinics are the mainstay	No consensus ; oxybutynin first-line if used	Antimuscarinic in the first year , same triggers	Start in the newborn if NDO is suspected
UTI	No prophylaxis on CIC ; intravesical antibiotics if recurrent	—	Under 3 years : positive urinalysis + positive catheter culture + fever $\geq 38.0^{\circ}\text{C}$. Over 3 years : add leakage between catheterisations and pelvic or back pain	Treat only if symptomatic
Botulinum toxin	Alternative for continence and UDS	Improves compliance	—	Refractory NDO
Surgery	Vesicostomy : temporary; lowers pressure, reverses hydronephrosis and VUR, reduces UTI; for those who cannot manage CIC. Augmentation : the definitive low-pressure reservoir ; cytology and endoscopy 5–10 years after . Continence : sling, bladder-neck reconstruction, artificial sphincter	Ileum preferred	Age ≥ 18 with an augmentation : cystoscopy and upper-tract imaging for tract change, gross haematuria , recurrent UTI, worsening incontinence, pelvic pain, renal transplant with BK polyomavirus	Augment the hostile bladder refractory to drugs and toxin ; channel if urethral CIC is difficult ; outlet surgery for sphincter weakness , usually with augmentation ; annual cystoscopy from 10–15 years after reconstruction

- **The table is the treatment ladder:**
 - **CIC plus antimuscarinic**;
 - then **botulinum toxin**;
 - then **augmentation with or without outlet surgery and a channel**.
- **SBA and EAU favour early, proactive treatment**; BAPU records no consensus.
- **Traps:**
 - **no antibiotic prophylaxis on CIC**;
 - **vesicostomy is a bridge, not definitive**.

GUIDELINE STANCE AT A GLANCE

Decision	Proactive	Selective or no consensus
Early CIC	SBA, EAU (ICCS if not emptying)	BAPU
Early antimuscarinic	EAU (newborn), SBA, ICCS	BAPU
Routine initial UDS	ICCS, SBA, EAU	BAPU

Medical Management

Clean Intermittent Catheterisation.

- **CIC is the single most important non-surgical tool.**
 - Regular emptying prevents injurious overdistension.
 - CIC bypasses DSD.
- **Early CIC is favoured. Families find it easier to learn and integrate CIC when it starts in infancy.**
- **If CIC fails to relieve hydronephrosis, increase capacity or reduce UTI → overnight indwelling drainage can help.**

Pharmacotherapy.

- **Antimuscarinics are first-line for overactivity or hypertonicity.**
- **Receptors:** the detrusor has M2 (more numerous) and M3 (which mediates contraction).
- **Response is good at first, but side effects and waning efficacy erode adherence.**
- **Effects of antimuscarinics:**
 - higher capacity;
 - higher volume at first contraction;
 - fewer incontinence episodes and catheterisations.
- **Side effects:** dry mouth, constipation, blurred vision, flushing, dizziness, headache.
- **A randomised double-blind trial found no effect on attention or memory.**

TABLE 46.5 PHARMACOTHERAPY FOR NEUROPATHIC LUT DYSFUNCTION

Drug	Class	Mechanism	Paediatric approval (US Food and Drug Administration [FDA])	Forms	Key paediatric dosing
Oxybutynin*	Antimuscarinic, tertiary amine	Mainly M3	≥5 years	Syrup, immediate-release (IR) and extended-release (ER) tablet, patch, gel, intravesical	Oral 0.1–0.6 mg/kg/day in 2–4 doses (0.2 mg/kg/dose up to three times daily), max 5 mg/dose; ER ≥6 years 5 mg daily, max 20 mg/day; intravesical 0.1–0.9 mg/kg/day, max 5 mg/dose; patch ≥4 years 3.9 mg/day, twice weekly
Tolterodine	Antimuscarinic, tertiary	Non-selective	Adult dosing	IR tablet, ER capsule	IR 2 mg twice daily; ER 4 mg daily
Solifenacin*			≥2 years		

Drug	Class	Mechanism	Paediatric approval (US Food and Drug Administration [FDA])	Forms	Key paediatric dosing
	Antimuscarinic, tertiary	M3; once daily		Suspension, tablet	Weight-banded, 2 mg (9–15 kg) to 5 mg (>60 kg); 3-weekly titration; max 4–10 mg/day
Fesoterodine*	Antimuscarinic, tertiary; prodrug	M3	≥6 years and >25 kg	ER tablet	4 mg daily → 8 mg; max 8 mg/day
Darifenacin	Antimuscarinic	—	Adult dosing	ER tablet	7.5 mg → 15 mg after ≥2 weeks
Trospium	Antimuscarinic, quaternary	Minimal central M1 affinity	Adult dosing	IR tablet, ER capsule	IR 20 mg twice daily; ER 60 mg each morning
Mirabegron*	β3 agonist	Adenylyl cyclase → cyclic AMP → detrusor relaxation	≥3 years (2021)	Granules, ER tablet	Granules by weight: 24 mg (11–22 kg), 32 mg (22–35 kg), 48 mg (≥35 kg); tablet 25 mg; titrate after 4–8 weeks (max 48/64/80 or 50 mg)
Vibegron	β3 agonist	As mirabegron	Not approved	IR tablet	Adult 75 mg daily
Ephedrine	α-sympathomimetic	Bladder-neck α-receptors → higher outlet resistance	Unlicensed for urinary use	Oral	0.5–1.0 mg/kg/day in three doses; ≥12 years 12.5–25 mg once or twice daily
Pseudoephedrine	α-sympathomimetic	As ephedrine	Unlicensed for urinary use	IR tablet	0.4 mg/kg twice daily to 0.9 mg/kg three times daily; 6–12 years 30 mg, ≥12 years 60 mg, once or twice daily

* FDA-approved for use in children; approved ages vary. Weight-based doses from the text; fixed age-band doses from the table.

- **The pattern:**

- **four agents are licensed for children: oxybutynin, solifenacin, fesoterodine, mirabegron;**
- **each has a child-friendly form or weight-banded dosing.**

- **The traps:**

- **vibegron is not licensed in children;**
- **sympathomimetics have no urinary licence;**
- **oral oxybutynin has a 5 mg per-dose ceiling.**

ANTIMUSCARINICS — DISTINGUISHING FEATURES AND SIGNATURE ADVERSE EFFECTS

Drug	Distinguishing feature	Signature adverse or practical issue
Oxybutynin	Commonest ; the only one with oral, transdermal and intravesical routes	Patch : skin irritation, about 20% stop. Intravesical : no first-pass metabolism , fewer side effects; low-level evidence; more stop because of inconvenience (crushing tablets) than side effects
Tolterodine	Reduces non-voiding activity and voiding amplitude	Poor brain penetration → minimal CNS effects ; capacity rises under 10 years and incontinence falls over 12 months
Fesoterodine	Prodrug with the same active metabolite (5-hydroxymethyl tolterodine)	Esterase activation , independent of cytochrome P450 → less pharmacokinetic variability and fewer adverse events than tolterodine
Trospium	Quaternary amine ; half-life about 8 hours versus 2 hours for oxybutynin	Crosses the blood-brain barrier poorly → few CNS effects ; adverse events uncommon
Solifenacin	Once daily ; from 2 years; suspension	Mostly mild; few stop
Propiverine	Mixed antimuscarinic and calcium antagonist	Urodynamic gain greater than with oxybutynin

- **β3 agonists:**

- **well tolerated;**
- **can cause hypertension;**
- mirabegron, **alone or with an antimuscarinic**, is effective in idiopathic and neuropathic overactivity;
- **first-line mirabegron may save costs.**

- **Propiverine:**

- **reduces calcium entry**, so there is **less calcium-calmodulin activation of the contractile apparatus;**
- **improves LUTS and above all urodynamics.**

Sympathomimetic Agents.

- **Indication:** incontinence from low urethral resistance.
- **Mechanism:** they act at the **bladder neck**, where **α-receptors are densest**, and raise outlet resistance.
- **Response:**
 - **only a minority respond**, and responders are **unpredictable;**
 - responders can become dry between catheterisations.
- **Adverse effects:**
 - **dizziness, nausea, nervousness, insomnia, anorexia, headache, mood change;**
 - **retention in children who void volitionally.**

Early Medical Intervention.

- **Definition:** early intervention = CIC plus antimuscarinic.
- **Indication:** proactive treatment of high-risk dynamics:
 - **DSD;**
 - **poor compliance;**
 - **high outlet pressure;**
 - **outflow obstruction.**
- **History:** introduced by Bauer in the 1980s. The field split into **proactive** and "wait and see" camps.

- **Why proactive management prevailed:**
 - the upper-tract risk of a hostile bladder was confirmed;
 - oxybutynin benefits the detrusor at cellular and muscarinic level.
- All paediatric urology societies now endorse proactive treatment of a hostile neonatal bladder.

EARLY INTERVENTION VERSUS EXPECTANT ("WAIT AND SEE") MANAGEMENT

Outcome	Expectant	Early CIC plus antimuscarinic
Bladder dynamics	Hostile dynamics persist	Contractions abolished or peak pressure lowered; filling pressure falls; minimal side effects
Continence	—	Improved; up to about 44% dry with normal upper tracts at 6 years
UTI	About 50% by 15 months, about 80% by 15 years; recurrence common	About 0.3 per year
Hydroureteronephrosis	18–80%	0–15%
VUR	Up to 50% by 9 years	Much less common
Diversion or augmentation	Diversion up to about 11%	18–24% fewer augmentations
Renal scarring	27–50%	4–12%
ESRD	About 30%	0–1.6%

- Early intervention gives no benefit to the child who empties spontaneously.
- Renal damage at birth: estimates range from 30% to 8% (UMPIRE) with DMSA defects. Either way, neuropathic LUT dysfunction harms the kidney.

KEY POINTS: NEUROPATHIC LOWER URINARY TRACT: MANAGEMENT

- CIC is the single most important non-surgical tool.
- Antimuscarinics are first-line, with no adverse cognitive effect (randomised double-blind trial).
- All paediatric urology societies recommend early intervention (CIC plus antimuscarinic) for hostile neonatal dynamics.

Concept — Early intervention works because it **attacks the cause (pressure and failure to empty) before the consequence (reflux, dilatation, scarring, renal failure)**. Each row of the comparison is the same mechanism measured at a different end-organ. The spontaneously emptying child, who gains nothing, proves the rule.

Surgical Management

- When medical treatment fails, surgery aims to:
 - improve dynamics;
 - achieve continence;
 - facilitate CIC.
- Augmentation, bladder-neck procedures and channels are covered in the reconstruction chapter.

Botulinum Toxin.

- **Mechanism:**
 - **botulinum toxin A (BTA)**, from *Clostridium botulinum* (Gram-positive), **blocks acetylcholine release**;
 - it **also blocks ATP and substance P release**;
 - it **down-regulates purinergic and capsaicin receptors on afferents**;
 - it **reduces M2, M3, P2X2 and P2X3 expression**.
 - It therefore acts on **both the motor and sensory arms**.
- **History:** first used in adult SCI (1990); in children in 2002.
- **Technique:** general anaesthesia → rigid cystoscopy → about 10 U/kg (max 360 U) → about 30 sites → trigone spared.
- **Outcomes after failed CIC plus drugs:**
 - **continence 65–87%**;
 - **lower detrusor pressure, higher capacity, better compliance**;
 - **effect lasts 6–10.5 months**;
 - usual **minimum age 2 years**, which matches the **FDA cerebral palsy licence**.
- **Wider benefits:**
 - **satisfaction >70%**;
 - **about 25% stop antimuscarinics after the first injection**, and about 10% more after the second;
 - **close to 90% avoid augmentation**.
- **Side effects are infrequent:** retention, UTI, haematuria, **transient weakness**, dysuria, pain.
- **Repeat injections:**
 - **stay effective, with lengthening intervals**;
 - **reducing the dose (300 → 200 U) caused relapse**, reversed by returning to the full dose;
 - **no tachyphylaxis, antibody effect or fibrosis**. Histology shows **fibrosis actually falls**.
- **Poor response:**
 - **predicted by pre-existing poor compliance**;
 - **complete non-response is linked to anti-toxin antibodies** (inconsistently).
- **Adding sphincter injection** (cohort with DSD in over 90%) **improved post-void residual**. This is preliminary.

Management of Vesicoureteric Reflux in Neuropathic Dysfunction of the Lower Urinary Tract.

- **Prevalence:** dilating VUR (grades III–V) is present in **7% of SB neonates**.
- It **almost always accompanies hostile dynamics** (poor compliance, overactivity, DSD). It is **rare without them**.
- **Untreated, it reaches 30–40% by 5 years**.

MANAGEMENT OF VUR BY PATTERN (INTERNATIONAL CLASSIFICATION)

Situation	Management
Grade I–III, and empties completely (voids spontaneously, or complete lesion with low outlet resistance)	Antibiotic prophylaxis alone
Grade IV–V	CIC
Cannot empty, any grade	CIC

Situation	Management
Poor compliance, with or without hydroureteronephrosis	Add an antimuscarinic

- Managed this way, VUR resolves in 30–55%.
- Bacteriuria on CIC is common but harmless unless VUR is high-grade.

CREDÉ MANOEUVRE — REACTIVE VERSUS DENERVATED SPHINCTER

Reactive external sphincter	Denervated, non-reactive sphincter
Credé → reflex rise in EMG and urethral resistance → high voiding pressure	No rise in EMG, resistance or pressure
Worsens VUR and its water-hammer effect: avoid	Not harmful

- Antireflux surgery — indications (as for a normal bladder):
 - recurrent febrile UTI despite prophylaxis and proper CIC;
 - persistent hydroureteronephrosis despite effective emptying and lowered pressure;
 - severe VUR with an anatomical abnormality of the ureterovesical junction.
- Jeffs (1976): reimplantation succeeds if complete emptying is ensured.
 - Before this, diversion was preferred.
 - Since CIC, success is about 95%.
- Unilateral VUR → unilateral surgery. Contralateral reflux does not follow.
- At augmentation:
 - low-grade VUR resolves without reimplantation;
 - grade ≥III persists in 20–47% after augmentation alone, so reimplant at the same operation.

OPEN REIMPLANTATION VERSUS ENDOSCOPIC INJECTION

Open ureteroneocystostomy	Endoscopic injection
About 84–96% success	About 61–73% in comparative studies; 73% (one) and 91% (two injections) in single-arm series; about 95% with intradetrusor BTA
Durable	About 40% recur at a mean of 4.5 years
Standard	Reasonable alternative; results reportedly similar to those in normal bladders

- **Why the endoscopic figures differ:** single-arm series use a **selected denominator**; head-to-head cohorts report lower success.

KEY POINTS: NEUROPATHIC LUT DYSFUNCTION — SURGICAL MANAGEMENT

- **Surgery is used when medical treatment fails:** to improve dynamics, achieve continence and facilitate CIC.
- **Intradetrusor BTA improves continence and UDS and may avoid reconstruction;** serial injection causes neither fibrosis nor loss of effect.
- **Dilating VUR is present in 7% of newborns with SB. Avoid Credé.**
- **Antireflux indications are as for a normal LUT.**

- **Reimplant dilating VUR during augmentation.**

Concept — Reflux in the neuropathic bladder is **a consequence of the bladder, not a primary ureteric defect**. Treat emptying and pressure first and much reflux resolves; surgery succeeds only on a bladder made safe. That is why Jeffs' insight and CIC turned reimplantation from failure to about 95% success.

SEXUAL FUNCTION

- **Sexual function matters increasingly as patients reach adulthood.** Nearly three-quarters of adults with SB report sexual interest.
- **Psychosocial barriers:**
 - parents and clinicians avoid the topic;
 - teenagers are **socially isolated** and learn less from peers;
 - dependence and parental overprotection limit autonomy.
- **Supports:**
 - men living independently report better erections and satisfaction;
 - parental permissiveness and age-appropriate treatment support self-esteem.
- **Dissatisfaction is highest for partnership and sex life** (more than half).
- **Central precocious puberty (CPP) is more common:**
 - approaching 50% in girls;
 - 10–30% in boys.
 - **Likely cause:** hydrocephalus disturbing the hypothalamic–pituitary–gonadal axis, together with Chiari II.
- **Education and contraception:**
 - offer SB-specific sexual education;
 - latex-free condoms are mandatory with latex allergy or avoidance;
 - combined oestrogen–progestin → **thromboembolism** with reduced mobility;
 - intrauterine device → **unsafe** (pelvic infection).
- **Determinants of sexual function:**
 - **function is inversely proportional to the neurological level;** the lowest lesions form partnerships most often;
 - more incontinence and disability → **less activity**;
 - older age → **more partners**;
 - no hydrocephalus → **more activity**, especially in women.
- **Women are more often sexually active but face sexual abuse** (about a third report unwanted attention or touching).
- **Teenagers:**
 - about a third have had sexual encounters;
 - almost all hope to marry and have children;
 - most desire sexual contact.
- **Studies exclude the severely disabled, so they overstate function.**

SEXUAL AND REPRODUCTIVE FUNCTION — MALES VERSUS FEMALES WITH SB

Males	Females
Sacral cord often involved → erection and ejaculation affected	Hormonal control → reproductive function spared
Erection: about 70–92% in older selected series; only about 40% normal in a later cohort; ambulation predicts it	Sexual dysfunction about 55% on the Female Sexual Function Index (FSFI); 89% in a combined FSFI and International Index of Erectile Function (IIEF) study
Ejaculation 40–75%; some retrograde	Continence → better lubrication and pain scores; shunt, mobility and CIC made no difference
S1 or lower → usually normal reproductive function; above S1 → about 50%	70–80% conceive with an uneventful pregnancy; late incontinence and caesarean delivery common
Fathering 17–39% (plus about 25% with a good prognosis); infertility from erectile dysfunction (ED), poor semen, Sertoli-cell-only testes	Arousal correlates with quality of life
IIEF, ≥18 years: about 75% ED, linked to a sacral root lesion; intact pelvic parasympathetic reflexes are protective	—

- Both sexes have more severe dysfunction than other causes, with similar desire.
- IIEF may miss young men's experience, so a disease-specific tool is needed.
- Offspring risk:
 - 3.7% if one parent has SB (higher if the mother);
 - about 15% if both.
- Folic acid for women with SB: 4–5 mg daily before contemplating pregnancy.
- Treatment:
 - sildenafil helps about 80% of men with myelomeningocele (randomised double-blind placebo-controlled trial);
 - ilioinguinal-to-dorsal penile nerve transfer (men with normal groin sensation) improves glans sensation and satisfaction in small series; long-term data awaited.

KEY POINTS: SEXUAL FUNCTION

- Sexuality matters increasingly as patients reach adulthood.
- CPP risk is increased.
- Function is inversely proportional to the neurological level.
- 70–80% of women conceive with an uneventful pregnancy; caesarean is commoner.
- 3.7% risk of SB in offspring.
- 75% of men ≥18 have ED; sildenafil is effective.
- 55.5% of women have sexual dysfunction.

Concept — The sexes diverge on *what the sacral cord controls*. Erection and ejaculation are sacral-neural and usually involved; female reproduction is hormonal and spared. So in men the lesion level sets function, with S1 as the watershed, whereas in women continence and psychosocial factors dominate.

OTHER CONDITIONS ASSOCIATED WITH NEUROPATHIC LOWER URINARY TRACT DYSFUNCTION

- **Conditions are grouped by brain, spinal cord and pelvis, whether congenital or acquired.**

Conditions of the Brain

Cerebral Palsy

- **Cerebral palsy (CP) is the commonest physical disability of childhood.**
 - **About 3.3 per 1000** eight-year-olds;
 - varies by region; commoner in Black and White than Hispanic children;
 - **male : female 1.2 : 1.**
- **Definition:** a disorder of **movement and posture from a non-progressive fetal or infant brain disturbance**. Sensation, cognition, communication and behaviour may also be affected.
- **Gross Motor Function Classification System — Expanded and Revised (GMFCS-E&R, 2007):**
 - based on **self-initiated movement**;
 - **five levels**, from **walks without limitation** to **transported in a manual wheelchair**;
 - age-banded.
- **Continence is usually achieved, but late:**
 - **daytime first, night-time within about a year**;
 - **14–34% continent** overall;
 - **median age 3.6–4.1 years** with good intellect and diplegia or hemiplegia; **10–13 years** with low intellect and tetraplegia.
- **GMFCS level mostly does not predict continence**, but **non-walkers have more LUT dysfunction** (about three-quarters versus one-quarter).
- **LUTS increase with age.**
- **Symptoms:**
 - **16–94%**, commoner in **girls**;
 - **incontinence commonest (mostly 35–45%)**;
 - urgency and frequency;
 - monosymptomatic enuresis 3–13%;
 - **UTI 5–57%**;
 - **constipation 33–66%**.
- **Upper-tract deterioration is under 5% (0.5–12%)**. It is mainly hydronephrosis; also asymmetry, VUR and microlithiasis.
 - **Risk factors: age, DSD symptoms (retention, interrupted voiding, hesitancy), recurrent UTI.**
- **Urodynamics:**
 - **normal in one-third**;
 - **overactivity 30–61%**;
 - underactivity about 6%;
 - **small capacity 42–93%**;
 - **DSD about 12%**;
 - raised residual about 13%.

- **Uroflow is mostly bell-shaped**; less often staccato, intermittent, plateau or tower.

Tumours/Infarcts/Encephalopathies

- **CNS tumours present mainly with incontinence and retention.** UTI, straining, hydronephrosis and urgency also occur.
- **New bladder symptoms may herald tumour progression or recurrence.**
- **Urodynamics:**
 - poor compliance, inability to void, high voiding pressure, overactivity, raised residual, abnormal sphincter;
 - **no correlation with tumour site** (intracranial versus extracranial, suprasacral versus sacral).
- **Newborn hypoxic-ischaemic encephalopathy (HIE):**
 - **lower voided volume and arousal with voiding;**
 - **higher residual and voiding frequency.**

Conditions of the Spinal Cord

Trauma

- **Paediatric SCI is rare:**
 - **Sweden about 2.6 per million per year;**
 - **USA: spinal-injury admissions about 170 per million under 21 years, SCI about 24 per million.**
- **Incidence rises with age:** about six times higher in adolescents than in children.
- **Sex: boys predominate, except in young children, where the sexes are equal.**
- **Causes:**
 - **motor vehicle or bicycle accidents (about a quarter to a half);**
 - falls, gunshot, diving or sport.
- **Cause by age:**
 - **infants:** vehicle;
 - **toddlers and children:** falls;
 - **adolescents:** sport.
- **Iatrogenic:** spinal deformity surgery, aortic surgery, **ductus arteriosus ligation.**
- **Neonatal: hyperextension during high forceps delivery.**
- **Why the child's cord is vulnerable** (hypermobility → **ischaemic necrosis**):
 - **horizontal facet joints** → anteroposterior subluxation;
 - **late paraspinal muscle and ligament support;**
 - **a heavy head placing maximal flexion in the upper cervical spine.**

SPINAL SHOCK VERSUS RECOVERED PHASE AFTER SCI

Spinal shock	After recovery
Underactive bladder, non-reactive sphincter; needle EMG potentials may look normal	Contractility and reactivity return as oedema settles
—	Suprasacral lesion (reticulospinal pathways cut) → overactivity with DSD

Spinal shock	After recovery
—	Cauda equina → little or no return

- **Sacral sensation and reflexes do not predict final LUT function.**
- **Thoracic lesion, long term: overactivity with DSD, high voiding pressure, then hydronephrosis and VUR.**
 - Typical pattern: **good compliance for part of filling, then small, C-fibre-mediated rhythmic contractions** with a **waxing and waning sphincter**.
 - **Leakage, but no effective emptying.**
- **Autonomic dysreflexia:**
 - **lesion at or above T6;**
 - **hallmark: severe hypertension**, which can cause **stroke, seizures or death**;
 - **mechanism:** loss of supraspinal control of splanchnic sympathetic outflow, so **bladder filling or contraction triggers an unopposed α 1-mediated sympathetic surge**;
 - **rule: monitor blood pressure and have α -blockers available during cystography and UDS.**

Transverse Myelitis

- **Transverse myelitis (TM) is an immune-mediated inflammatory myelopathy.**
- **Epidemiology:**
 - **about 1400 US cases a year, 28% in children;**
 - **incidence peaks at 10–19 years and in the third decade;**
 - **sexes equal.**
- **Diagnostic criteria:**
 - **cord-attributable sensory, motor or autonomic dysfunction;**
 - **bilateral signs;**
 - **a defined sensory level;**
 - **inflammation (cerebrospinal fluid pleocytosis or increased T2 MRI signal).**
- **Presentation: sudden back pain or leg weakness → rapid paralysis → retention.**
 - **Bladder dysfunction usually follows the motor deficit.**
 - **Bowel dysfunction is likely.**
- **Course:**
 - **spinal shock up to 6 weeks**, then spasticity if recovery is incomplete;
 - **the cord segment involved determines the motor deficit.**
- **Acute treatment: steroids, plasma exchange, intravenous immunoglobulin.**
- **Recovery: begins at 2 weeks–3 months; mostly within 6 months; may continue up to 2 years.**
- **Retention in more than 95% during the acute phase.**

TRANSVERSE MYELITIS — ACUTE VERSUS POST-SHOCK URODYNAMICS

Acute phase	After spinal shock
Areflexia or underactivity about 70%	Overactivity 59–90%
Overactivity about 13%	Poor compliance about 47%
Poor compliance about 30%	DSD 17–80%

Acute phase	After spinal shock
DSD about 20%	DLPP >40 cm H ₂ O in 12–33%; underactivity rare

- **Outcomes:**
 - about one-third void spontaneously;
 - 57–73% need CIC;
 - antimuscarinics in 14–64%;
 - BTA or augmentation if refractory;
 - bowel dysfunction persists in up to about three-quarters.
- **Motor and bladder recovery usually move together:**
 - full → full;
 - partial → voids but needs an antimuscarinic;
 - none or wheelchair-dependent → CIC with or without an antimuscarinic.
- **Complications are infrequent:**
 - UTI, VUR, hydronephrosis;
 - chronic renal insufficiency;
 - calculi, diverticula.
- **Late CIC → poorer compliance and upper-tract deterioration (about 5%).**
- **Examination, symptoms and ambulation do not predict upper-tract risk.**
- **Largest series:**
 - intact leg reflexes and full recovery → better volitional voiding;
 - but more than half on CIC still need it at 1 year.

Conditions of the Pelvis

Anorectal Malformation

- **Anorectal malformation (ARM):**
 - distal anorectal anomalies, with urinary and genital involvement;
 - about 1 in 5000 live births;
 - Krickenbeck (International) Classification by fistula site.
- **Urinary anomalies occur in 26–52%, chiefly left renal agenesis and VUR. Boys are affected more often than girls (about 50% versus 29%).**

HIGH VERSUS LOW ANORECTAL MALFORMATION

High (supralelevator fistula)	Low (infralevator)
Urinary anomaly about 70%	About 35%
Spinal bony anomaly 48–54%	15–27%
Cord anomaly more frequent (overall 18–50%)	Less frequent

- **Spinal bony anomalies overall: 30–44%.**
- **Cord anomalies: tethered cord, thick or fatty filum, lipoma.**
 - Neuropathic LUT dysfunction is frequent with a cord anomaly, rare without one;
 - it usually shows as incontinence when toilet training fails.
- **Skin markers are rarely present, so spinal ultrasound before 3 months is needed.**

SYNDROMIC ANORECTAL MALFORMATIONS

Syndrome	Components
VACTERL	Vertebral, Anal atresia, Cardiac, Tracheo-oEsophageal fistula, Renal, Limb
MURCS	Müllerian duct aplasia, Renal aplasia, Cervicothoracic Somite dysplasia
OEIS	Omphalocele, Exstrophy, Imperforate anus, Spinal defects
Currarino	ARM + sacrococcygeal defect + presacral mass

- **Currarino syndrome:**
 - tethered cord in more than 80%;
 - about 80% have postoperative voiding complaints;
 - upper-tract changes are not uncommon;
 - untethering has unpredictable effects on urodynamics.
- **Urodynamics:**
 - commonest: upper motor neuron pattern (overactivity and/or DSD);
 - lower motor neuron pattern possible;
 - leak point pressure >40 cm H₂O, poor compliance, small capacity;
 - normal urodynamics do not exclude a vertebral anomaly or SB.
- **ARM aetiology:**
 - multifactorial and partly genetic;
 - associations: paternal smoking, maternal obesity and diabetes.
- **The neuropathic bladder in ARM comes from the cord, not the surgery:**
 - Peña's posterior sagittal anorectoplasty (PSARP) spares the pelvic nerves;
 - 18–35% have neuropathic dysfunction;
 - nearly all abnormal before PSARP, usually with a cord anomaly.
- **Urodynamics in ARM:**
 - indications: spinal bony anomaly, cord defect, or VCUG or ultrasound signs;
 - timing: early infancy before definitive surgery → after the pull-through → again if secondary urinary or faecal incontinence develops;
 - an abnormal infant study → cord surgery or watchful waiting.
- **ESRD just under 2%.**
- **If neuropathic → annual ultrasound plus repeat UDS or imaging (Table 46.3).**

Pelvic Surgery

- **Pelvic operations that harm LUT function:**
 - sacrococcygeal teratoma resection;
 - pelvic rhabdomyosarcoma and other pelvic tumours;
 - Hirschsprung disease surgery.

PELVIC NERVE INJURY AND LUT PATTERN

Nerve injured	Autonomic loss	LUT result
Pelvic splanchnic	Parasympathetic	Underactive bladder
Hypogastric	Sympathetic	Loss of compliance; bladder neck unresponsive to abdominal pressure

Nerve injured	Autonomic loss	LUT result
Pelvic plexus	Mixed	Mixed

PELVIC SURGERY — THREE CONDITIONS COMPARED

	Sacroccygeal teratoma	Pelvic rhabdomyosarcoma / tumours	Hirschsprung disease
LUT dysfunction	35–45% neuropathic	Normal in only 27–40%	Unstable detrusor up to 45%, often resolving; transient retention up to 7%; incontinence 0–6%
Presentation	UTI, incomplete emptying	Enuresis, daytime incontinence, urgency, suprapubic pain on filling, UTI	Usually silent
Urodynamics	Normal 28%; overactivity 57–81%; underactivity 14–27%; poor compliance; DSD 18–38%	Markedly small capacity; over- or underactivity; DSD	One-third normal; two-thirds large capacity; high residual 55–78%; poor compliance rare
Upper tract	Scarring or low GFR about 25%; VUR 25–50%; hydronephrosis 30–43%	Abnormal renal function one-third; hydronephrosis 15%; reconstruction or diversion about 20%	—
Special point	Late tethering may contribute	Cystectomy then; bladder preservation now; radiotherapy → high risk	Surgical approach affects outcome
Screening	UDS before and after surgery in all	Frequency-volume chart for all survivors; UDS and ultrasound only if abnormal	Clinical screen and ultrasound before and after surgery for all; UDS before surgery if symptomatic, and 6 months after surgery if symptomatic (neurapraxia resolves)

- Long-term follow-up in all three: annual ultrasound and review if UDS is abnormal or symptoms persist.
- Asymptomatic children with normal postoperative tests need no long-term follow-up unless symptoms arise.

OTHER CONDITIONS AT A GLANCE

Condition	Dominant UDS pattern	Upper-tract risk	Key point
CP	Overactivity, small capacity; one-third normal	Low (<5%)	Risk rises with age, DSD symptoms and UTI
CNS tumour	Mixed; no site correlation	Variable	New symptoms may mean recurrence
SCI	Shock → suprasacral: overactivity with DSD; cauda equina: no return	High (thoracic)	Autonomic dysreflexia at or above T6
TM	Acute areflexia → overactivity, DSD, poor compliance	About 5%; higher with late CIC	Examination does not predict risk; UDS needed
ARM	Upper motor neuron; lower motor neuron possible	ESRD about 2%	Spinal ultrasound before 3 months; UDS before and after pull-through

Condition	Dominant UDS pattern	Upper-tract risk	Key point
Pelvic surgery	Depends on the nerve injured	High after teratoma or rhabdomyosarcoma	Screening as tabulated

KEY POINTS: OTHER CONDITIONS ASSOCIATED WITH NEUROPATHIC LOWER URINARY TRACT DYSFUNCTION

- **CP: continence is usually achieved, but late;** upper-tract deterioration is **under 5%**.
- **SCI is rare, and six times commoner in adolescents than in children.**
- **Spinal shock: underactive bladder, non-reactive sphincter.**
- **Upper thoracic or cervical SCI → autonomic dysreflexia. Monitor blood pressure and have α -blockers available** during cystography and UDS.
- **TM: bladder dysfunction usually follows the motor deficit.** Reflexes and full recovery predict voiding, but **more than half on CIC still need it at 1 year.**
- **ARM: cord anomalies in 18–50%.**
- **Monitor pelvic rhabdomyosarcoma survivors closely, especially after radiotherapy.**

Concept — Across the non-NTD causes, **the level of the lesion predicts the pattern:** suprapontine (CP, brain) gives overactivity with coordination and low renal risk; suprasacral cord (SCI, TM, tethering) gives overactivity with DSD and high renal risk; sacral or peripheral (cauda equina, pelvic surgery) gives underactivity or denervation. **In ARM and pelvic surgery, the cord or nerve lesion, not the visceral operation, is usually the culprit.**

FIGURE LEGENDS

- **Fig. 46.2 (sacral agenesis, from L5):**
 - the lower end of the cord is **squared off at T11;**
 - the **sacroiliac joints lie in the midline;**
 - **no vertebral bodies below L4.**
- **Fig. 46.6: a tethered cord after myelomeningocele repair, with the conus at L3–L4.**
- **Fig. 46.7: infant spinal ultrasound (arches unossified) showing an intradural lipoma displacing the cord anteriorly from L3, confirmed on MRI.**
- **Fig. 46.9: on oxybutynin, the uninhibited contraction is delayed to a much larger volume at lower pressure.**
- **Fig. 46.10: ephedrine raises urethral pressure at the bladder neck, with little change at the external sphincter.**
- **Fig. 46.13: TM, with a T2 cord signal from T3 to T8.**
- **Fig. 46.14: imperforate anus with vertebral anomalies:**
 - **tethered cord;**
 - **bilateral hydronephrosis;**
 - **left VUR;**
 - **right ureterovesical junction obstruction;**
 - **poor compliance and dyssynergia.**